University of Ottawa

CSI 3140 - WWW Structures, Techniques and Standards

Laboratory 1

Web Foundations, HTML5 Structure, Links, Forms, and Accessibility-Aware Markup

Professor	Mohamed Ali Ibrahim, PhD, PEng
Term	Spring/Summer 2026
Lab Period	Monday, May 11, 2026 to Saturday, May 23, 2026
Submission	Brightspace, by Saturday, May 23, 2026, 11:59 PM
Team Size	2-3 students
Technologies	HTML5, basic CSS for readability, browser developer tools, W3C/HTML validation tools

1. Laboratory Overview

The purpose of this laboratory is to introduce students to the foundations of the World Wide Web and to give them hands-on practice with modern, standards-compliant HTML5. Students will create a small multi-page website that demonstrates correct document structure, hyperlinks, forms, and accessibility-aware markup.

This laboratory is aligned with the beginning of the course, including Web foundations, clients and servers, Internet protocols, HTTP, HTML document structure, semantic markup, accessibility basics, links, images, tables, forms, and client-side validation basics.

2. Topics Covered in Lab 1

Topic Area	Required Coverage
Web foundations	Clients, servers, browsers, URLs, DNS, HTTP request/response basics, and the role of Web standards.
HTML5 document structure	DOCTYPE, html, head, metadata, title, body, header, nav, main, section, article, aside, and footer.
Links and navigation	Internal links, external links, relative paths, email links, anchors, and consistent navigation across pages.
Content markup	Headings, paragraphs, lists, images with alt text, simple tables with captions and headers.
Forms	Labels, inputs, text areas, select menus, radio buttons, checkboxes, buttons, required fields, placeholder text used appropriately, and basic HTML5 validation attributes.
Accessibility-aware markup	Semantic HTML, meaningful link text, alternative text, heading hierarchy, explicit labels, keyboard-friendly structure, and basic ARIA only when necessary.
Validation and testing	Use browser developer tools and an HTML validator to check syntax and structure.

3. Learning Objectives

- Explain the basic roles of clients, servers, browsers, URLs, DNS, and HTTP in Web communication.
- Create a valid HTML5 document with a clear and semantic structure.
- Build a small multi-page website using relative links and consistent navigation.
- Create accessible links, images, tables, and forms using appropriate HTML elements and attributes.

- Use form elements and HTML5 validation attributes to collect structured user input.
- Apply basic accessibility principles, including semantic structure, alt text, labels, and logical heading order.
- Validate HTML pages and fix common structural errors.
- Prepare a short technical report explaining the design, the structure of the website, accessibility choices, and validation results.

4. Problem Description

Each team must design and implement a small informational website for a fictional university event, club, workshop, or technical mini-course. The website must be built using HTML5 and must demonstrate Web foundations, semantic structure, links, forms, and accessibility-aware markup.

The website does not need to be visually complex. The main objective is to produce correct, meaningful, standards-based HTML. Basic CSS may be used only to improve readability, but the grading emphasis is on HTML structure, accessibility, forms, links, and validation.

5. Required Website Structure

The submitted website must contain at least four HTML pages:

Page	Required Content
index.html	Home page introducing the website topic. Must include semantic sections and navigation links to all other pages.
about.html	Information page explaining the event, club, workshop, or topic. Must include headings, paragraphs, lists, and at least one image with appropriate alt text.
schedule.html	Schedule or program page. Must include a table with a caption, column/row headers, and meaningful structure.
registration.html	Registration or contact page. Must include an accessible HTML form with labels and HTML5 validation attributes.

All pages must include consistent navigation and must be connected using relative hyperlinks. Each page must use a meaningful title and a logical heading hierarchy.

6. Part A - Web Foundations Summary

In the lab report, each team must include a short explanation of the Web communication process used when a browser loads one of their pages. The explanation should mention:

- The role of the browser as a client.
- The role of a Web server.
- The purpose of a URL.
- The basic role of DNS.
- The idea of an HTTP request and HTTP response.
- The difference between local files during development and files served by a Web server.

7. Part B - HTML5 Semantic Structure

Each HTML file must use a correct HTML5 structure. The following skeleton illustrates the expected structure:

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Page Title</title>
</head>
<body>
  <header>
    <h1>Website Title</h1>
    <nav aria-label="Main navigation">
      <a href="index.html">Home</a>
      <a href="about.html">About</a>
      <a href="schedule.html">Schedule</a>
      <a href="registration.html">Registration</a>
    </nav>
  </header>

  <main>
    <section>
      <h2>Section Title</h2>
      <p>Page content goes here.</p>
    </section>
  </main>

  <footer>
    <p>&copy; 2026 CSI 3140 Lab 1</p>
  </footer>
</body>
</html>
```

8. Part C - Links, Images, Lists, and Tables

Students must demonstrate correct use of links and common HTML content elements.

Required Elements

- At least one internal navigation link on every page.
- At least one external link to a relevant resource.
- At least one email link or contact link.
- At least one image using a meaningful alt attribute.
- At least one ordered or unordered list.
- At least one table with caption, thead or appropriate header cells, and readable content.

9. Part D - Accessible HTML Form

The registration.html page must include a form. The form does not need to submit data to a server, but it must be structurally correct and accessible.

Minimum Form Requirements

- At least six input fields or controls.
- Each input must have an explicit label connected with the for and id attributes.
- Include at least one text input, one email input, one select menu, one radio group or checkbox group, one textarea, and one submit button.
- Use HTML5 validation attributes such as required, minlength, maxlength, type="email", pattern, or min/max where appropriate.

- Group related form controls using fieldset and legend when useful.
- Avoid using placeholder text as the only label.

```
<form action="#" method="post">
  <fieldset>
    <legend>Participant Information</legend>

    <label for="full-name">Full name</label>
    <input type="text" id="full-name" name="fullName" required minlength="2">

    <label for="email">Email address</label>
    <input type="email" id="email" name="email" required>

    <label for="program">Program of study</label>
    <select id="program" name="program" required>
      <option value="">Select a program</option>
      <option value="computer-science">Computer Science</option>
      <option value="software-engineering">Software Engineering</option>
      <option value="computer-engineering">Computer Engineering</option>
    </select>
  </fieldset>

  <button type="submit">Submit registration</button>
</form>
```

10. Part E - Accessibility and Validation Checklist

Each team must complete the following checklist and include it in the report.

Item	Completed?
Each page has exactly one main h1 heading.	
Headings follow a logical order without skipping levels unnecessarily.	
Navigation links use meaningful text.	
Images have meaningful alt text or empty alt text only when decorative.	
Tables include captions and header cells.	
Form controls have explicit labels.	
The form can be navigated using the keyboard.	
The HTML files were checked using a validator.	
Major validation errors were fixed.	
The pages were tested in at least one modern browser.	

11. Laboratory Demonstration Requirement

Each team must demonstrate the laboratory work to the TA. The report will be considered for grading only if the demonstration has been completed. All team members must be present during the demonstration.

During the demonstration, the TA may ask any team member to explain the website structure, navigation, form design, accessibility choices, validation results, and the role of each submitted file. Each student must be able to explain the team submission clearly.

12. Testing Requirements

Each team must test the website before submission. The testing must include:

- Opening each HTML file in a modern browser.
- Checking that all internal navigation links work correctly.
- Checking that all images load and include appropriate alt text.
- Testing the form with valid and invalid input.
- Checking that required fields trigger browser validation.
- Testing keyboard navigation through the main links and form controls.
- Validating each HTML page using an HTML validator and saving evidence of the results.

13. Deliverables

Each team must submit one ZIP file on Brightspace containing:

- All HTML files: index.html, about.html, schedule.html, and registration.html.
- Any CSS file used for basic readability, if applicable.
- Any image files used in the website.
- A short lab report in PDF format.
- Screenshots showing the website pages in a browser.
- Screenshots or exported results showing HTML validation for each page.

14. Report Requirements

The lab report must include:

- Course code, lab number, team members, and student numbers.
- A short description of the website topic.
- A file structure diagram or list explaining all submitted files.
- A short explanation of the Web communication process.
- A description of the HTML5 semantic elements used.
- A description of the form design and validation attributes used.
- The completed accessibility and validation checklist.
- Screenshots of the website pages and validation results.
- A short reflection on problems encountered and how they were solved.

15. Grading Rubric

Component	Points
Correct multi-page website structure and file organization	10
HTML5 semantic structure and valid document skeleton	15
Correct links, navigation, paths, images, lists, and table markup	15
Accessible and well-structured HTML form with validation attributes	20
Accessibility-aware markup: headings, labels, alt text, table headers, meaningful links, and keyboard-friendly structure	15
Validation, browser testing, and evidence of corrections	10
Laboratory demonstration to the TA with all team members present	10
Report quality, clarity, screenshots, and reflection	5
Total	100

16. Important Notes

- Late submissions will not be accepted unless an official accommodation or approved exceptional circumstance applies.
- All team members must understand the submitted website and report.
- External code, templates, images, or resources must be cited if used and must be permitted for academic use.
- Students must not submit a website generated entirely by an external tool without understanding, adapting, and documenting the work.
- Academic integrity rules apply to all submitted files and reports.

17. Suggested Timeline

Date	Suggested Work
Monday, May 11	Read the lab statement, form teams, and choose the website topic.
May 12-14	Create the file structure and implement the main HTML5 skeleton for all pages.
May 15-17	Add navigation, links, images, lists, and the schedule table.
May 18-20	Implement the registration form and accessibility improvements.
May 21	Validate all HTML files and fix errors.
May 22	Complete screenshots, report, and final testing.
Saturday, May 23	Submit the ZIP file on Brightspace by 11:59 PM.